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Genetic overlap of chronic pain, musculoskeletal-specific pain, substance use disorders and substance use consumption: Common addiction and substance-specific effects

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Abstract

Background and aims: Chronic pain and substance use disorders frequently co-occur and may share general addiction-related and substance-specific genetic pathways. Additionally, substance consumption and problematic use may differ in their genetic associations with pain subtypes. We applied genomic structural equation modeling to assess shared genetic and neurological enrichment between general and musculoskeletal-specific chronic pain to general addiction liability, substance use disorder-specific risk and substance use.

Design: Cross-trait genomic structural equation modeling using genome-wide association study (GWAS) summary statistics from large-scale global consortia.

Setting: Data were aggregated from international research collaborations and biobanks in the United States and Europe.

Participants: Participants were individuals (n = 63 982 to 2 428 851) with European-like ancestry from population-based and clinical cohorts.

Measurements: We analyzed a General Chronic Pain factor from 24 chronic pain GWASs and a Musculoskeletal-specific Pain factor from 11 overlapping conditions. We constructed an Addiction factor from GWASs of alcohol, tobacco, cannabis and opioid use disorders, and partitioned out substance-specific genetic risk. We used alcohol and smoking consumption GWASs to distinguish use from problematic use. We applied molecular neurological annotations to assess brain cell type enrichment in shared genetic effects.

Findings: The Addiction factor was genetically correlated with General Chronic Pain [genetic correlation (r_g)_ = 0.448, false discovery rate (FDR)-corrected $P < 0.001$, 95% confidence interval (95% CI) = 0.40, 0.49], but not with Musculoskeletal-specific Pain (r_g = 0.001, FDR-corrected P = 0.998, 95% CI = −0.07 to 0.07). Controlling for Addiction, General Pain had substance-specific associations: negative correlations with Problematic Alcohol Use (r_g = −0.155, FDR-corrected P = 0.049, 95% CI = −0.28 to −0.02) and Alcohol Frequency (r_g = −0.334, FDR-corrected $P < 0.001$, 95% CI = −0.46 to −0.21), and a positive correlation with Cigarette Frequency (r_g = 0.274, FDR-corrected $P < 0.001$, 95% CI = 0.21, 0.34). Musculoskeletal-specific Pain positively associated with Tobacco Use Disorder (r_g = 0.186, FDR-corrected P = 0.020, 95% CI = 0.05, 0.32) and Alcohol Frequency (r_g = 0.161, FDR-corrected P = 0.049, 95% CI = 0.02, 0.30). General Pain related to Cannabis Use Disorder and Opioid Use Disorder entirely through the Addiction factor.

Shared genetic components were enriched in excitatory and inhibitory neurons, glia and endothelial cells.

Conclusions: General chronic pain appears to share substantial genetic overlap and neuronal enrichment with addiction risk and shows distinct associations with problematic alcohol use, alcohol consumption and cigarette use, reflecting both a broad transdiagnostic mechanism as well as substance-specific paths. In contrast, musculoskeletal-specific pain appears to be more narrowly linked to tobacco use disorder and alcohol use frequency, but not to general addiction liability.

KEYWORDS
addiction, chronic pain, functional enrichment, genomics, musculoskeletal pain, substance use

INTRODUCTION

Chronic pain and substance use disorders (SUDs) are highly co-prevalent conditions that present a significant public health burden [1–3]. Chronic pain, defined as pain persisting for 3+ months [4], affects more than 30% of people worldwide [5]. Substance use and disordered use contribute over 4% to the global burden of disease and disability [6]. The intersection of substance use and chronic pain carries broad social consequences. For instance, the prescription of opioids for pain treatment fueled the opioid epidemic and subsequent abuse of other drugs [7, 8]. Prescription opioid misuse occurs in up to 44% of patients with chronic pain [9], and medical cannabis, which is increasingly being used to manage chronic pain, has been accompanied by rising rates of cannabis use disorder [10]. Moreover, more than 20% of individuals with chronic pain smoke cigarettes [1, 11], and up to 38% of people who binge drink report using alcohol for pain relief [12]. Yet, the extent to which SUDs are associated with pain through shared versus substance-specific etiological pathways remains unclear.

Chronic pain and SUDs share genetic liability [13–15]. Latent genetic factors have been identified for SUDs, capturing common variance and separating out smaller, substance-specific variance [13, 16, 17]. The common, transdiagnostic addiction factor [13] is a genetic factor that reflects mechanisms shared across SUDs, such as dysregulation in dopaminergic pathways and reward-processing circuits [18], mechanisms also implicated in pain chronification [11, 19–21]. Substance-specific genetic variance captures unique biological mechanisms specific to the substance, such as variation in receptors, transporters or enzymes (e.g. alcohol dehydrogenase activity in alcohol use) [17, 18]. Understanding how pain genetically associates with common addiction and substance-specific variation may elucidate the pathophysiology of pain and its co-occurrence with SUDs. For example, smoking may share specific genetic risk with pain owing to individual variation in the expression of central and peripheral nicotinic acetylcholine receptors [11], while also relating to pain through variation in common dopaminergic pathways [11, 19, 21].

Furthermore, substance use has separable genetic variance from SUDs, which may differentially relate to pain. For example, alcohol use and alcohol use disorder are only modestly genetically correlated (r_g = 0.33) [22]. Genes such as *KLB* are specific to alcohol use, while others, like *ADH1B*, *ADH1C* and *DRD2*, are shared with problematic use [23]. Partitioning substance use and disordered use in relationship to pain can inform etiological paths; for example, alcohol consumption is associated with enriched gene sets for cellular responses to alcohol drinking, whereas problematic alcohol use is associated with significant enrichment in the post-synaptic modulation of chemical transmission [23].

Chronic pain involves peripheral and central (e.g. cognitive and affective) processing [24–26]. In addition to etiologies unique to specific pain conditions, phenotypic [27] and genetic [28–30] studies have identified common factors underlying various pain conditions, irrespective of body site. A genetic general pain factor constructed from 24 pain conditions reflects central, affective pain mechanisms, as it shows selective genetic expression in the central nervous system and genetic overlap with cognitive, mood, sleep and personality traits, psychiatric disorders, neuroplasticity and inflammation regulation, implicating the nervous system [30, 31]. An orthogonal musculoskeletal-specific pain genetic factor was constructed from the additional covariance of 11 musculoskeletal pain conditions, beyond the covariance explained by the general pain factor. This factor has gene sets and pathways associated with skeletal development, transcription machinery, choline transcription machinery and genetic overlap with anthropomorphic traits [30]. Importantly, prior work examining pain and substance use associations has not isolated musculoskeletal-specific pain variance while controlling for general pain variance, which would enable an examination of differential associations with substance use phenotypes for this subset of pain conditions.

We used genomic structural equation modeling (GenomicSEM) [32] to assess the degree of genetic relatedness of the general pain and the musculoskeletal pain factors with the addiction factor, and with substance use and SUDs over and above common addiction liability. We examined the degree of functional enrichment of brain cell types of significant genetic associations to inform biological pathways. We expected to find shared genetic risk between general pain and addiction; however, the relationships with other SUDs, consumption traits and the functional enrichments tests are exploratory. This approach informed the level of associations and common and unique biological pathways shared between chronic pain and substance use.

METHODS

Summary statistics

We obtained summary statistics from previously published genome-wide association studies (GWASs). In a GWAS, a trait of interest is regressed on each single nucleotide polymorphism (SNP) across the genome to obtain SNP effect sizes, which can be used as summary statistics for subsequent analyses. In other words, GWASs identify which genetic variants are associated with the trait of interest, such as number of drinks consumed per week. The summary statistics we used were limited to European ancestry given the insufficient sample sizes for non-European ancestries and that GenomicSEM requires analyses to be performed within ancestry. For the models described below, all indicators are GWAS summary statistics for the corresponding phenotypes. Descriptive statistics for all GWASs can be found in Table 1.

Chronic pain GWASs

The summary statistics for the 24 pain conditions were from GWASs of data from the UK Biobank [30]. Chronic pain was defined as pain lasting longer than 3 months or conditions with persistent pain as an important symptom, such as osteoarthritis.

Substance use GWASs

The GWAS summary statistics for opioid use disorder (OUD) [33], problematic alcohol use (PAU) [34], Alcohol Use Disorder Identification Test—Consumption (AUDIT-C) items [22], drinks per week (DPW) [35], tobacco use disorder (TUD) [36], cigarettes per day (CPD) [35] and cannabis use disorder (CanUD) [37] were included. Case definition varied for the SUDs, but generally a case was indicated by the presence of an International Classification of Diseases (ICD) or Diagnostic Statistical Manual (DSM) substance use disorder code. Controls were mostly defined by an absence of an ICD or DSM substance use disorder code. For OUD, the controls were a mix of participants who were: exposed, screened and did not meet diagnosis; unexposed, screened and did not meet diagnosis; and unscreened but did not have an ICD/DSM code for OUD. PAU is a continuous measure that includes alcohol use disorder (AUD), alcohol dependence (AD) and alcohol-related problems identified in questions 4–10 of the AUDIT—Problem test. Definitions for continuous traits (AUDIT-C, CPD, DPW and PAU) were scales measuring problematic use or consumption frequency.

Measurement models in GenomicSEM

Chronic pain and addiction models

We fitted measurement models for the chronic pain and addiction factors using GenomicSEM (Appendix S1). The chronic pain model is a bi-factor-like model with a general pain factor that captures variance of all 24 pain conditions and an additional orthogonal musculoskeletal-specific pain factor that captures variance of 11 musculoskeletal conditions, over and above the general pain factor. The addiction factor is a common factor model that reflects the shared variance of all four substance use disorder traits (i.e. OUD, PAU, TUD and CanUD); this factor is an update of the 2022 addiction model using the most recent, well-powered GWASs of SUDs [13].

Alcohol and tobacco Cholesky decompositions

We fitted Cholesky decompositions, which is a statistical modeling approach that partitions the shared and unique variance across traits in a sequence [38]. Each latent factor explains variance in the first trait, and its shared variance with all subsequent traits (e.g. factor 1 captures all of the variance of trait 1 and the variances of traits 2 and 3 that are shared with trait 1; factor 2 captures the residual variance of trait 2 not explained by factor 1 and the variance in trait 3 shared with that residual variance; factor 3 captures the residual variance of trait 3 that is not explained by factors 1 and 2). This approach partitions variance to identify how much overlap exists between traits, and whether later traits have unique influences that are not shared with earlier ones. The alcohol Cholesky decomposition included three factors: factor 1, the all alcohol use factor, which captured all variance of PAU and variance of AUDIT-C and DPW shared with PAU; factor 2, the alcohol consumption residual factor, which captured residual variance of AUDIT-C that was not shared with PAU and the variance of DPW shared with this residual variance in AUDIT-C; and factor 3, the alcohol frequency (DPW) residual factor. The tobacco Cholesky decomposition included two factors: factor 1, the all tobacco use factor, which captured all variance of TUD and variance of CPD shared with TUD; and factor 2, the tobacco frequency (CPD) residual factor. For Cholesky decompositions, all indicator residual variances are fixed to 0 so that the factors fully captured the variances [38]. We did not include opioid and cannabis consumption, because there are no currently publicly available well-powered GWASs for these behaviors.

GenomicSEM analyses

SNP heritability and co-heritability of substance use and chronic pain traits.

We estimated the SNP heritability (proportion of phenotypic variance explained by genetic variance) of four SUDs, three substance consumptions and 24 chronic pain conditions from the GWAS summary statistics mentioned earlier, using linkage disequilibrium score regression (LDSC) in the GenomicSEM R package [39]. LDSC estimates the heritability of a trait and genetic correlations between traits using GWAS summary statistics. As GWASs are often meta-analyzed across multiple cohorts, there is often sample overlap in traits; LDSC controls for these sampling dependencies. Then, also in

TABLE 1 Descriptive statistics of GWAS summary statistics.

Phenotype	Full name	Source	Cohorts	SNP h^2	Sample	Prevalence	Case/definition	Control
arth	Arthritis	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.09	238 195	0.34	Presence of diagnosis code	Absence of code
back	Back pain	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.09	251 857	0.47	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months
chDs	Chest pain/ discomfort	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.08	431 571	0.17	Presence of diagnosis code	Absence of code
chPh	Chest pain during physical activity	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.13	63 982	0.05	Presence of diagnosis code	Absence of code
crpl	Carpal tunnel	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.16	435 971	0.03	Presence of diagnosis code	Absence of code
CWP	Chronic widespread pain	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.14	433 905	0.01	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months
cyst	Cystitis	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.03	204 624	0.08	Presence of diagnosis code	Absence of code
enLL	Enthesopathies of lower limb	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.06	202 713	0.04	Presence of diagnosis code	Absence of code
enth	Enthesopathies	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.06	203 831	0.14	Presence of diagnosis code	Absence of code
gast	Gastritis	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.07	221 716	0.19	Presence of diagnosis code	Absence of code
gout	Gout	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.20	207 322	0.07	Presence of diagnosis code	Absence of code
hdch	Headache	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.13	386 144	0.10	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months
hipA	Hip arthrosis	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.14	210 724	0.08	Presence of diagnosis code	Absence of code
hipP	Hip pain	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.08	422 962	0.10	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months
IBS	Irritable bowel syndrome	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.07	211 295	0.13	Presence of diagnosis code	Absence of code
kneA	Knee arthrosis	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.14	216 030	0.15	Presence of diagnosis code	Absence of code
kneP	Knee pain	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.10	413 319	0.19	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months
legP	Leg pain	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.10	149 725	0.28	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months

TABLE 1 (Continued)

Phenotype	Full name	Source	Cohorts	SNP h^2	Sample	Prevalence	Case/definition	Control
mgrn	Migraine	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.12	211 460	0.10	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months
nksh	Neck/shoulder pain	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.08	402 144	0.18	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months
oesp	Oesophagitis	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.06	208 332	0.06	Presence of diagnosis code	Absence of code
rhAt	Rheumatoid arthritis	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.08	206 810	0.04	Presence of diagnosis code	Absence of code
pnjt	Pain in joint	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.05	435 971	0.03	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months
stmP	Stomach pain	Zorina-Lichtenwalter *et al.* 2023	UKBB	0.08	417 533	0.05	Presence of diagnosis code and 3+ months	Absence of code/less than 3 months
OUD	Opioid use disorder	Deak *et al.* 2022	MVP, PGC, iPSYCH, FinnGen, Partners Biobank, BioVU, Yale-Penn3	0.13	554 186	0.03	ICD or DSM OUD code	Unscreened, unexposed and no diagnosis
PAU	Problematic alcohol use	Zhou *et al.* 2023	MVP, FinnGEN, UKBB, PGC, iPSYCH, QIMR, Yale-Penn3,	0.06	903 147		AUD, alcohol dependence and AUDIT problem scale	–
TUD	Tobacco use disorder	Toikumo *et al.* 2023	UKBB, BioVU, MGBB, PMBB, MVP	0.11	739 895	0.24	≥1 code of ICD TUD code	Absence of a TUD code
CanUD	Cannabis use disorder	Levey *et al.* 2023	PGC/deCODE, MVP, iPSYCH2, MGBB, Yale-Penn3	0.06	886 025	0.05	≥1 code of in-patient or outpatient visit for CanUD	Absence of a CanUD code
AUDIT-C	AUDIT consumption	Sanchez-Roige *et al.* 2018	UKBB	0.11	121 604		Items 1–3 from AUDIT, consumption	–
DPW	Drinks per week	Saunders *et al.* 2022	GSCAN2	0.04	2 428 851		Drinks per week	–
CPD	Cigarettes per day	Saunders *et al.* 2022	GSCAN2	0.08	618 489		Cigarettes per day	–

Note: The descriptive statistics of each GWAS summary output file and its source are displayed above. SNP h^2 shows the proportion of a trait's phenotypic variance that can be explained by the additive genetic variation of SNPs identified in a GWAS. Prevalence is given for case/control phenotypes. Abbreviations: BioVU = Vanderbilt University Medical Center biobank; deCODE = deCODE genetics; FinnGen = Finnish Genomics Initiative; iPSYCH = The Lundbeck Foundation Initiative for Integrative Psychiatric Research; GSCAN = GWAS & Sequencing Consortium of Alcohol and Nicotine Use; GWAS = genome wide association study; MGBB = Mass General Brigham Biobank; MVP = Million Veteran Program; Partners Biobank = Partners Healthcare Biobank; PGC = Psychiatric Genetics Consortium; PMBB = Penn Medicine Biobank; QIMR = Queensland Institute of Medical Research Biobank; SNP h^2 = single nucleotide polymorphism heritability; UKBB = UK Biobank; Yale-Penn3 = Yale-Penn Cohort 3rd release.

LDSC, we estimated all pairwise genetic covariances, standardized to genetic correlations using each condition's SNP heritability as variance.

Structural models of genetic overlap between substance use and chronic pain factors.

Using GenomicSEM, we fitted a series of models based on prior literature [13, 30] to investigate the following questions.

1. To what degree do general chronic pain and musculoskeletal pain relate to general addiction liability?
   We ran a model correlating the general pain and musculoskeletal pain factors with the addiction factor.
2. Do these pain factors relate specifically to OUD, PAU, TUD and CanUD, when controlling for addiction liability?
   We ran partial association models to test the degree and significance of the relationships between each addiction indicator (OUD, PAU, TUD and CanUD) and the general pain and musculoskeletal

pain factors, while controlling for the addiction–pain correlations. The significant paths from each partial association model were included in one model to see which common and substance-specific associations persist when accounting for all associations.

3. To what degree do the chronic pain factors associate with substance use, when controlling for SUD-specific and common addiction liability?

   We used Cholesky decompositions to partition alcohol/tobacco problematic use and consumption variance. These Cholesky decompositions were built on top of the common and substance-specific association model described for question 2 (e.g. for the alcohol Cholesky decomposition, any substance-specific associations with the three other SUD indicators were still included in the model). We assessed genetic correlations between the pain factors and alcohol/tobacco problematic use and substance consumption, controlling for the addiction factor. For example, a significant association between alcohol frequency and pain represents the relationship pain has to alcohol frequency when controlling for common addiction and for problematic alcohol use.

Stratified GenomicSEM analyses

Estimating functional enrichment for model parameters

Stratified GenomicSEM uses a stratified LDSC matrix to assess functional enrichment of model parameters (e.g. factor correlation between general pain and addiction) that were estimated in GenomicSEM. Stratified LDSC is similar to LDSC, but functional information is used to partition SNP heritability into biological categories [40]. A functional annotation is considered enriched for the parameter of interest (e.g. factor variance) when the genetic variance of that annotation is greater than the proportional size of the annotation [41]. We used 113 annotations for the analysis, including: (i) baseline annotations, which include all SNP-level annotations (e.g. minor allele frequency bin); and (ii) neuronal subtype annotations (e.g. GABAergic neurons) [41, 42]. The full list of annotations is found in Appendix S2. We applied the 'enrich' function to test functional annotation of the significant parameters of interest. We applied stratified GenomicSEM to significant parameters from the common and substance-specific associations model as well as from the disorder/consumption Cholesky decompositions.

Inference criteria and model evaluation

Model goodness of fit was assessed by multiple fit indices; in general, a comparative fit index (CFI) > 0.90 and a standardized root mean squared residual (SRMR) < 0.08 are considered acceptable model fits for GenomicSEM models [32]. Significance, valence and effect size of the model parameters were informative of how chronic pain risk relates to substance use based on the parameterization of the model. Chi-square difference tests were conducted to assess whether dropping a parameter leads to a deterioration in the fit ($P < 0.05$) for nested models. We applied the false discovery rate (FDR) correction to each set of parameters tested for a model, as it provides greater power and appropriate error control for correlated genomic tests compared with Bonferroni correction, with a significance threshold defined as FDR-corrected $P < 0.05$ [43]. The analysis was not pre-registered and the results should be considered exploratory.

Software and code availability

The analyses were conducted in R 4.5.0 [44]. We used version 0.0.5 of GenomicSEM [32]. Data pre-processing and analysis code are publicly available on GitHub (https://github.com/lydia-rader/sud_pain_gsem_2025).

RESULTS

Descriptive statistics

The genetic correlation matrix is presented in Figure 1. Clusters in a genetic correlation matrix indicate clusters of shared genetic variance that may be well represented by a genetic factor. This correlation matrix showed a cluster of SUDs, the clustering of chronic pain conditions, and some negative clustering between substance use consumption and pain conditions. Descriptive correlations between the pain factors and each substance trait are presented in Table 2. Most substance use and pain conditions were positively correlated. The alcohol frequency measures were negatively correlated with many pain conditions (Figure 1). The bi-factor model of chronic pain [$\chi^2(237) = 1289.338$, CFI = 0.947, SRMR = 0.075] and the addiction factor model [$\chi^2(2) = 6.300$, CFI = 0.998, SRMR = 0.033] both fit well.

Common addiction and substance-specific genetic associations with pain

The individual substance-specific models (Table 3) were used to inform the parameterization of the model visualized in Figure 2. The addiction factor showed significant genetic overlap with general chronic pain ($r_g = 0.448$, FDR-corrected $P < 0.001$, 95% CI = 0.40, 0.49). The addiction factor did not significantly correlate with musculoskeletal pain ($r_g = 0.001$, FDR-corrected $P = 0.998$, 95% CI = −0.07, 0.07).

General pain had a negative PAU-specific association ($b = -0.105$, FDR-corrected $P = 0.020$, 95% CI = −0.19, −0.02), which is in the opposition direction of the full correlation (Table 2), suggesting a suppression effect. Musculoskeletal pain had a positive TUD-specific association ($b = 0.135$, FDR-corrected $P = 0.004$, 95% CI = 0.05, 0.22). Even

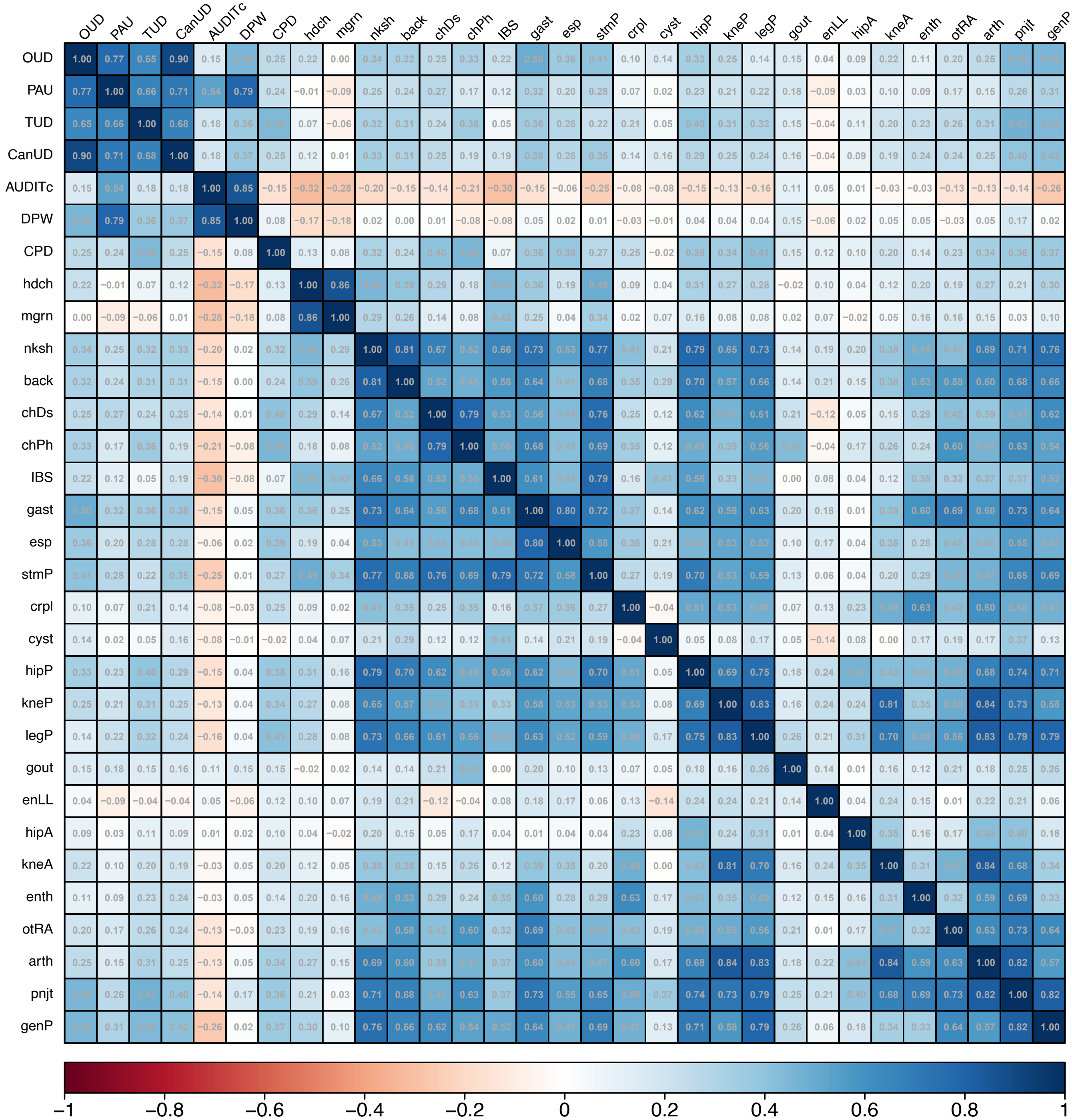

FIGURE 1 Linkage disequilibrium score regression (LDSC) matrix of genetic correlations. The LDSC matrix of genetic correlations between all substance use and chronic pain condition traits. OUD = opioid use disorder; PAU = problematic alcohol use; TUD = tobacco use disorder; CanUD = cannabis use disorder; AUDITc = AUDIT consumption score; DPW = drinks per week; CPD = cigarettes per day; hdch = headache; mgrn = migraine; nksh = neck/shoulder; back = back; chDs = chest pain/discomfort; chPh = chest pain during physical activity; IBS = irritable bowel syndrome; gast = gastritis; esp = esophagitis; stmP = stomach pain; crpl = carpal tunnel; cyst = cystitis; hipP = hip pain; kneP = knee pain; legP = leg pain; gout = gout; enLL = enthesopathies of lower limb; hipA = hip arthrosis; kneA = knee arthrosis; enth = enthesopathies; otRA = rheumatoid arthritis; arth = arthropathies, non-specific, including osteoarthritis; pnjt = pain in joint; genP = chronic widespread pain.

though musculoskeletal pain had a PAU-specific association in the individual substance models, this association was not significant (b = −0.029, FDR-corrected P = 0.540, 95% CI = −0.10, 0.04) in the final model including the path to TUD. The pain factors were not specifically significantly associated with CanUD or OUD over and above the addiction factor (Table 3).

TABLE 2 Descriptive genetic correlations between chronic pain factors and substance use traits.

	General chronic pain		Musculoskeletal-specific pain	
Substance use traits	r_g (SE)	P	r_g (SE)	P
Opioid use disorder (OUD)	**0.413** (0.05)	**<0.001**	−0.032 (0.05)	0.563
Problematic alcohol use (PAU)	**0.290** (0.03)	**<0.001**	−0.028 (0.03)	0.389
Tobacco use disorder (TUD)	**0.355** (0.03)	**<0.001**	**0.128** (0.04)	**<0.001**
Cannabis use disorder (CanUD)	**0.370** (0.03)	**<0.001**	0.027 (0.04)	0.475
AUDIT consumption score (AUDIT-C)	**−0.252** (0.03)	**<0.001**	0.080 (0.04)	0.052
Drinks per week (DPW)	−0.004 (0.02)	0.876	**0.067** (0.03)	**0.030**
Cigarettes per day (CPD)	**0.386** (0.02)	**<0.001**	**0.130** (0.03)	**<0.001**

Note: Descriptive correlations between the general chronic pain and musculoskeletal pain factors and the substance use traits. Entries in bolded indicate significance at P < 0.05. r_g = genetic correlation.

TABLE 3 Estimates of common and specific pathway partial association models.

	Addiction–general pain		Addiction–MSK pain		Specific–general pain		Specific–MSK pain	
Model	r_g (95% CI)	FDR	r_g (95% CI)	FDR	b (95% CI)	FDR	b (95% CI)	FDR
OUD	**0.410 (0.37, 0.45)**	0.254	0.050 (−0.01, 0.11)	0.138	0.055 (−0.06, 0.17)	0.428	−0.076 (−0.19, 0.04)	0.254
PAU	**0.447 (0.40, 0.49)**	**0.031**	**0.068 (0.01, 0.13)**	**0.049**	**−0.107 (0.04, 0.19)**	**0.030**	**−0.089 (−0.16, −0.02)**	**0.031**
TUD	**0.398 (0.35, 0.44)**	**0.004**	−0.007 (−0.06, 0.05)	0.876	0.066 (−0.01, 0.14)	0.151	**0.132 (0.05, 0.21)**	**0.004**
CanUD	**0.418 (0.38, 0.46)**	0.844	0.043 (−0.01, 0.10)	0.191	−0.005 (−0.09, 0.08)	0.912	−0.012 (−0.09, 0.06)	0.844

Note: The associations between the addiction factor and the general chronic pain and the musculoskeletal-specific pain represent the common pathway that all four substances (e.g. alcohol, cannabis, opioid and tobacco) relate to the pain factors. The specific pathway is the relationship that each substance use disorder has individually with the pain factors while controlling for the common addiction association with the pain factors. Entries in bold indicate significance at P < 0.05.
Abbreviations: b = the beta path estimate; CanUD = cannabis use disorder; FDR = false discovery rate-corrected P-value; MSK = musculoskeletal; OUD = opioid use disorder; PAU = problematic alcohol use; r_g = the genetic correlation estimate; TUD = tobacco use disorder.

Pain associations with alcohol/tobacco use factors, controlling for common addiction and alcohol/tobacco problematic use factors

The alcohol Cholesky decomposition fitted acceptably and is shown in Figure 3 [χ^2(378) = 6568.047, CFI = 0.927, SRMR = 0.071]. The all alcohol use Cholesky factor shared a negative genetic relationship with general pain (r_g = −0.155, FDR-corrected P = 0.049, 95% CI = −0.28, −0.02), but not with musculoskeletal pain (r_g = −0.052, FDR-corrected P = 0.463, 95% CI = −0.18, 0.08). The alcohol consumption residual factor showed a negative relationship with general pain (r_g = −0.334, FDR-corrected P < 0.001, 95% CI = −0.46, −0.21) and a positive relationship with musculoskeletal pain (r_g = 0.161, FDR-corrected P = 0.049, 95% CI = 0.02, 0.30). The alcohol frequency residual factor was not significantly associated with general pain (r_g = 0.083, FDR-corrected p = 0.463, 95% CI = −0.11, 0.28), nor with musculoskeletal pain (r_g = 0.138, FDR-corrected P = 0.328, 95% CI = −0.09, 0.36).

The tobacco Cholesky decomposition fitted acceptably and is shown in Figure 4 [χ^2(353) = 4265.927, CFI = 0.939, SRMR = 0.071). The all tobacco use Cholesky factor did not significantly relate to general pain (r_g = 0.039, FDR-corrected P = 0.463, 95% CI = −0.06, 0.14), but did relate to musculoskeletal pain (r_g = 0.186, FDR-corrected P = 0.019, 95% CI = 0.05, 0.32). The tobacco consumption residual factor shared a significant relationship with general pain (r_g = 0.274, FDR-corrected P < 0.001, 95% CI = 0.21, 0.34), but not with musculoskeletal pain (r_g = 0.070, FDR-corrected P = 0.189, 95% CI = 0.05, 0.32).

Functional enrichment of associations

We applied stratified GenomicSEM to significant associations identified in GenomicSEM. The main results are summarized below, but all results can be found in Appendixes S1 and S2.

We tested the three significant parameters in the addiction and substance-specific association model. The addiction factor association with general chronic pain, the PAU-specific association with general pain and the TUD-specific association with musculoskeletal pain had 58, 52 and 17 significant annotations that met FDR-corrected significance, respectively (Figure 5). The associations largely showed significant enrichments across neurological annotations including: GABAergic neurons; endothelial cells; oligodendrocyte precursor cells (OPCs); oligodendrocytes; microglia; astrocytic

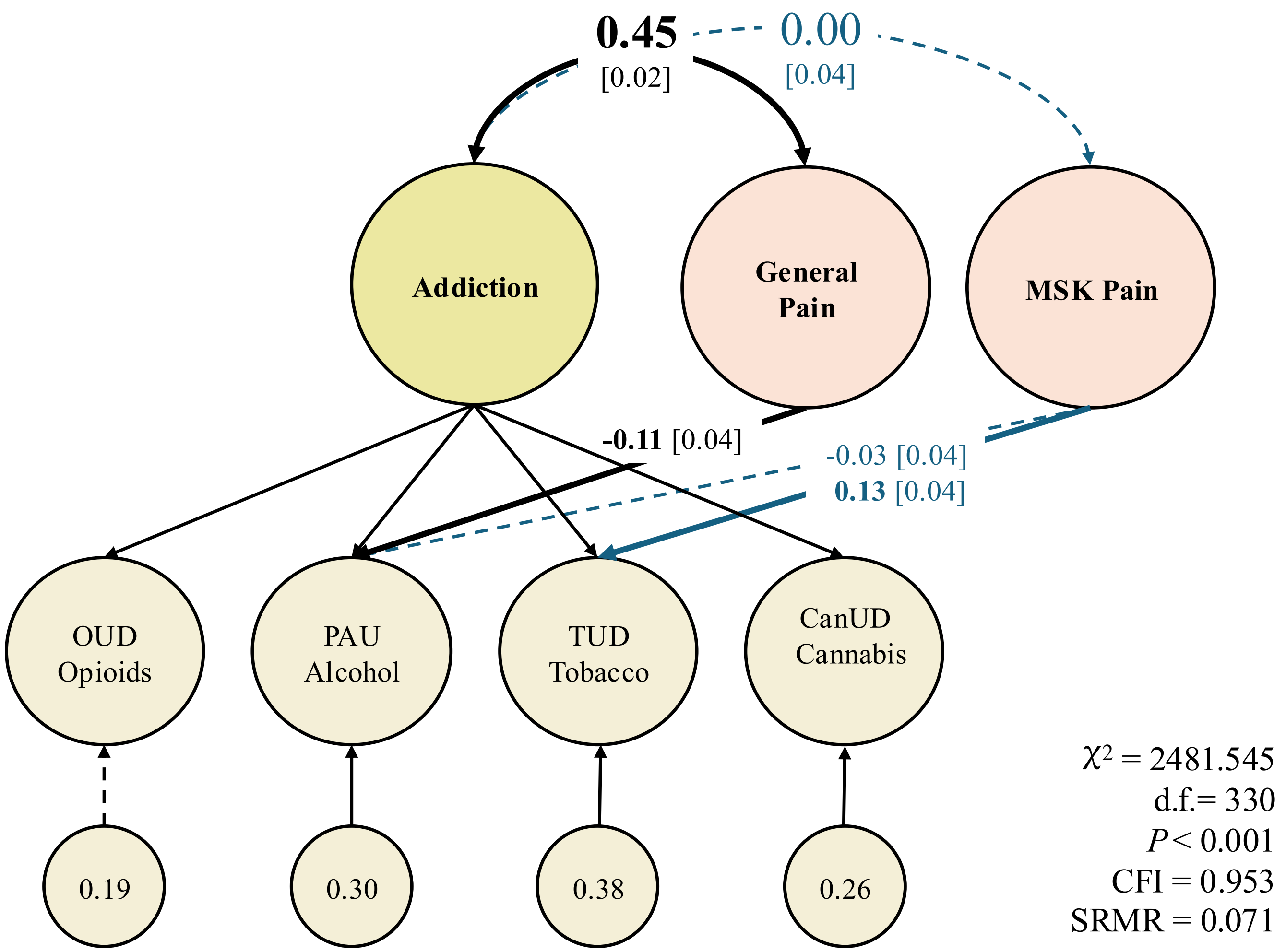

FIGURE 2 Common addiction and substance-specific associations with general pain and with musculoskeletal pain. In this model, we present the common paths from addiction to general pain and musculoskeletal pain, as well as the substance-specific associations that were significant when tested in the individual substance path models (Table 2). In the problematic alcohol use-only model, musculoskeletal pain had a significant association with problematic alcohol use, but this was no longer significant in this final model when including the tobacco use disorder path. The 24 pain indicators and their loadings are not visualized in this diagram, but the loadings can be found in Appendix S1. The correlations in bold font indicate the parameter is significant at a false discovery rate (FDR)-corrected $P < 0.05$ threshold. Standard errors are displayed in the closed brackets. CanUD = cannabis use disorder; MSK = musculoskeletal; OUD = opioid use disorder; PAU = problematic alcohol use; TUD = tobacco use disorder; χ^2 = chi-square statistic; d.f. = degrees of freedom; CFI = comparative fix index; SRMR = standardized root mean square residual.

transporters; excitatory CA1, CA3 and dentate gyrus hippocampal neurons; excitatory prefrontal cortex neurons; and neuronal stem cells (all FDR-corrected $P < 0.05$). Some neurological subtype enrichments occurred specifically on protein-truncating invariant genes (PI genes).

We tested three significant parameters in the alcohol Cholesky decomposition (Figure 3). The association between all alcohol use and general pain had 56 significant enrichments that passed FDR correction, with enrichment across neuronal subtypes. The alcohol consumption residual–general pain and the alcohol consumption residual–musculoskeletal pain associations both had zero significant enrichments, even prior to FDR correction. Neither of the two significant parameters from the tobacco Cholesky decomposition had enrichments that passed FDR correction.

DISCUSSION

General chronic pain shared substantial genetic risk and neurological enrichment with common addiction liability. Additionally, general chronic pain had substance-specific associations with problematic alcohol use, alcohol use and cigarette frequency. Musculoskeletal-specific pain showed genetic correlations with tobacco use disorder and alcohol frequency, but not with addiction.

Substance-general associations

General chronic pain shared a substantial genetic correlation with addiction, suggesting that the comorbidity between central pain

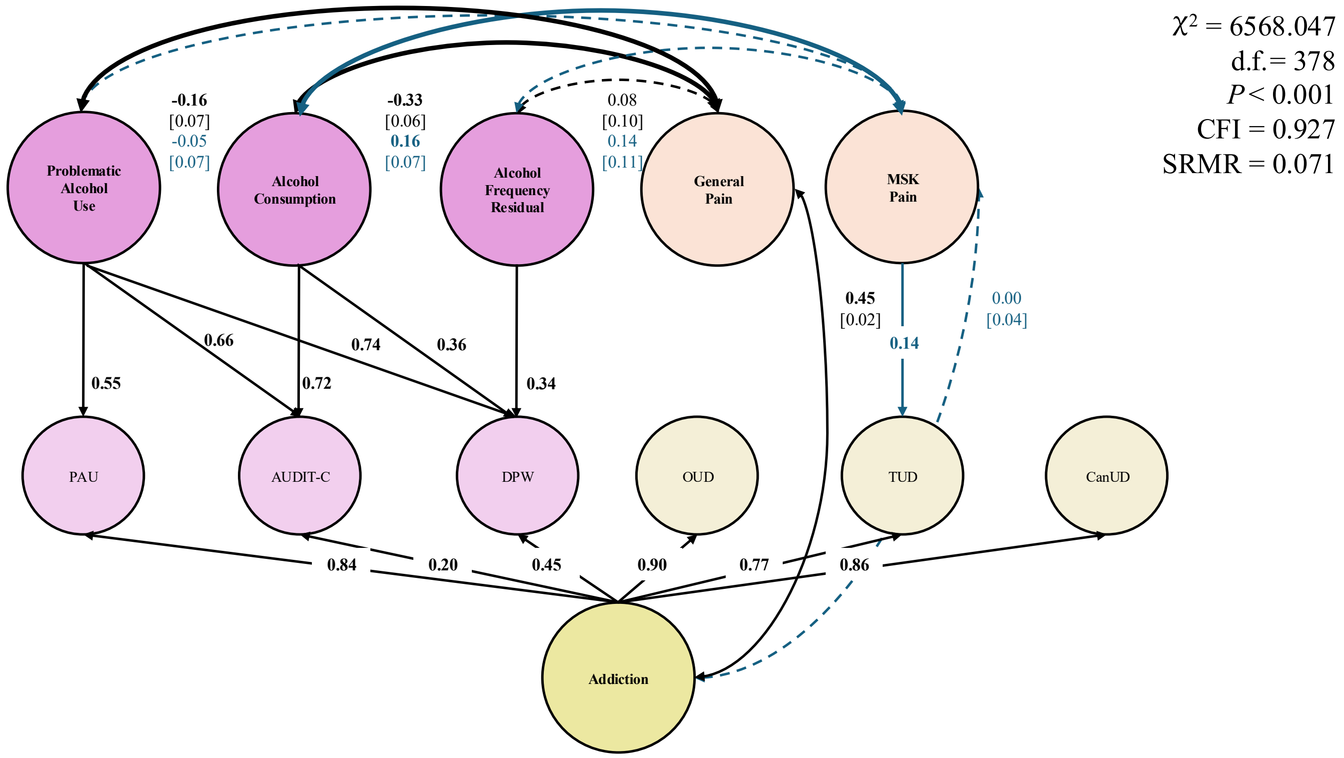

FIGURE 3 Genetic Cholesky decomposition of all alcohol use, alcohol consumption and relationships with chronic pain factors. In this model, we separate problematic alcohol use and alcohol consumption variance from each other, controlling for common addiction variance overall. Next, we assess the degree to which problematic alcohol use and alcohol consumption relate to the chronic pain factors. The PAU, AUDIT-C and DPW summary statistics were indicators on the addiction factor, and their circles represent the variance left over not accounted for by the addiction factor. This is why the PAU indicator loads onto the all alcohol use factor at 0.55 rather than at 1.00. The additional path to the TUD indicator from musculoskeletal pain was included in the model, based on the substance-specific findings found in Figure 2. The 24 pain indicators and their loadings are not visualized in this diagram but can be found in Appendix S1. The correlations in bold font indicate the parameter is significant at a false discovery rate (FDR)-corrected $P < 0.05$ threshold. Standard errors are displayed in the closed brackets. AUDIT-C = AUDIT consumption score; CanUD = cannabis use disorder; DPW = drinks per week; MSK = musculoskeletal; OUD, opioid use disorder; PAU = problematic alcohol use; TUD, tobacco use disorder; χ^2 = chi-square statistic; d.f. = degrees of freedom; CFI = comparative fix index; SRMR = standardized root mean squared residual.

mechanisms and SUDs is driven by a general addiction liability. This association could reflect horizontal pleiotropy, in which shared genetic variants contribute to the predisposition of multiple traits. Pleiotropic variants have been identified for multi-site chronic pain, SUDs and general addiction risk [14], supporting the existence of partially shared genetic predisposition.

The general chronic pain–addiction association was functionally enriched across GABAergic, glutamatergic and glial cell types, suggesting multiple, shared mechanisms. Specifically, a subcluster of GABAergic neurons were implicated in hippocampal CA1 and a subset of glutamatergic neurons were implicated in the prefrontal cortex [42]. As these cell subtypes were defined based on the similarity of their expression, their specific function as related to other neuronal cells of the same class remains equivocal. However, the involvement of GABAergic and glutamatergic neurons in the prefrontal cortex and hippocampus aligns with previous literature implicating disrupted inhibitory–excitatory balance in both central sensitization of chronic pain and addiction vulnerability [45–49]. It is likely that subgroups of long-range GABAergic neurons could implicate long-range (from dorsolateral prefrontal cortex to hippocampus) inhibitory processes in the shared pathophysiology of pain and addiction. Behaviorally, long-range communication between the increased engagement of emotional and motivational circuits coupled with aberrant GABAergic and glutamatergic signaling is known to be associated with the transition from acute to chronic pain, and similarly with the transition from substance use to misuse [49, 50]. Our analysis implicates subsets of GABAergic neurons in CA1, and more work should be done to identify whether this cluster represents a source of long-range communication between the dorsolateral prefrontal cortex and the hippocampus.

We also observed enrichment in astrocytic transporters, oligodendrocytes and OPCs, suggesting chronic neuroinflammatory processes and, more specifically, the potential role of altered synaptic plasticity [51]. In particular, OPCs are known to specifically respond to neuronal activity, and likely play a role in altering neuronal function via repeated stimulus [51]. These results are also in line with the aberrant glutaminergic processes implicated in our enrichment analysis, as OPCs can sense neuronal activity through their AMPA and NMDA receptors, and their proliferation and differentiation are activity dependent,

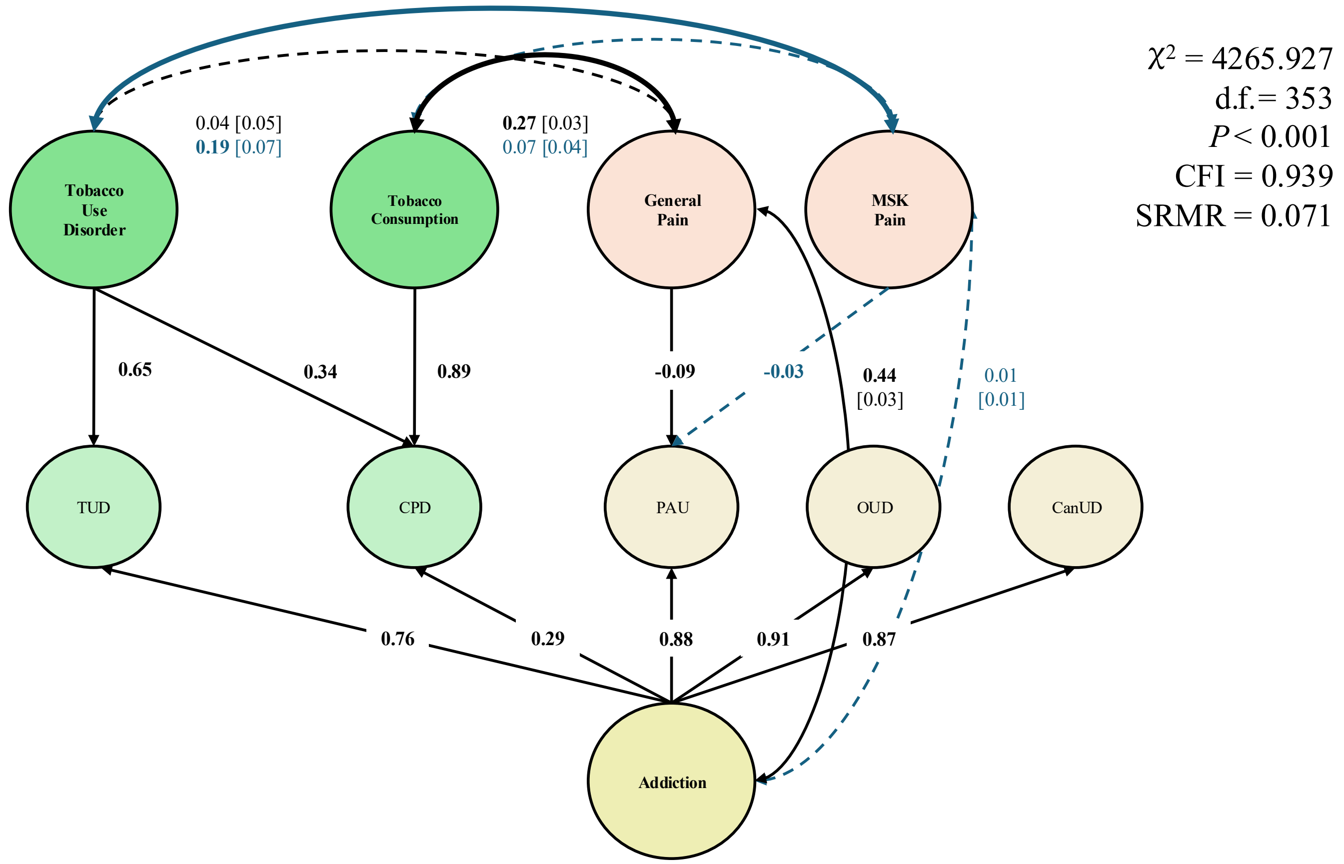

FIGURE 4 Genetic Cholesky decomposition of all tobacco use, tobacco consumption and relationships with chronic pain factors. In this model, we separate tobacco use disorder and tobacco consumption variance from each other, controlling for common addiction variance overall. Next, we assess the degree to which TUD and tobacco consumption relate to the chronic pain factors. The TUD and CPD summary statistics were indicators for the addiction factor, and their circles represent the variance left over not accounted for by the addiction factor. This is why the TUD indicator loads onto the all tobacco use factor at 0.65 rather than at 1.00. The additional paths from the PAU indicator to general pain and musculoskeletal pain were included in the model, based on the substance-specific findings found in Figure 2. The 24 pain indicators and their loadings are not visualized in this diagram but can be found in Appendix S1. The correlations in bold font indicate that the parameter is significant at a false discovery rate (FDR)-corrected $P < 0.05$ threshold. Standard errors are displayed in closed brackets. CPD = cigarettes per day; CanUD = cannabis use disorder; DPW = drinks per week; MSK = musculoskeletal; OUD, opioid use disorder; PAU = problematic alcohol use; TUD, tobacco use disorder; χ^2 = chi-square statistic; d.f. = degrees of freedom; CFI = comparative fix index; SRMR = standardized root mean squared residual.

allowing them to participate in learning-induced circuit remodeling and adaptive myelination, which optimizes neural network function [52]. The process of OPCs influencing white matter may influence the development and management of pain and SUDs. Exogenous OPC transplantation is being explored clinically for alleviating neuropathic pain by promoting re-myelination, and could potentially be explored for patients who are additionally experiencing substance misuse [51]. Our research provides additional evidence in support of the role of OPCs in chronic pain–substance misuse overlap and may extend the relevance of these cells to chronic pain conditions beyond neuropathic pain.

Additional enrichment in neuronal stem cells suggests that neural differentiation in the perinatal environment could confer risk for both chronic pain and addiction [53]. Furthermore, brain endothelial cell enrichment raises the possibility of blood–brain barrier dysfunction as a shared etiological pathway [54]. Taken together, our findings suggest that liability to multi-cellular dysfunction underlies the shared risk between general chronic pain and addiction.

These dysfunctions may reflect a heritable predisposition to altered neuronal signaling, receptor expression or glial responses. However, some of the association may be linked to vertical pleiotropy, where genetic variants influence one trait that has a causal effect on a second trait. In addition to discovering pleiotropic variants, Koller *et al.* [14] used Mendelian randomization to find evidence of bidirectional effects between multi-site chronic pain and alcohol, cannabis and tobacco. Their results suggest that multi-site chronic pain and opioid, alcohol, and cannabis use disorders occur through a mix of horizontal and vertical pleiotropy, or shared predisposition and causal pathways. Their findings suggested tobacco use disorder in particular may involve bidirectional associations with pain, rather than shared predisposing effects. However, Farrell *et al.* [55] analyzed eight pain conditions and found that smoking had a causal effect on back, neck/shoulder, hip, knee, abdominal and widespread pain, but not on facial pain or headaches, suggesting that causal relationships may be specific to individual pain conditions, rather than to general chronic pain

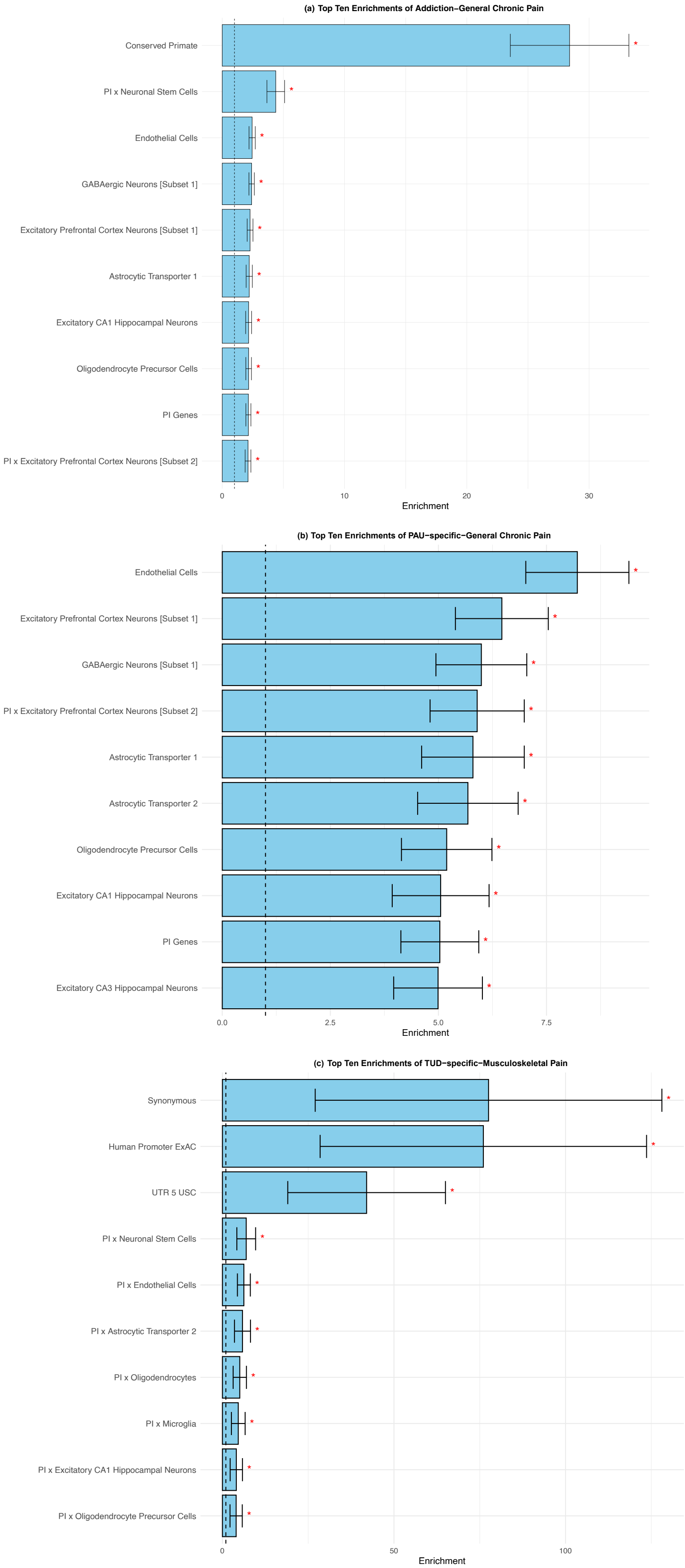

FIGURE 5 The most significant enrichments of the parameters in the common addiction and substance-specific association model. The top 10 significant functional enrichments for three parameters from the common addiction and substance-specific association model: (a) addiction with general pain; (b) PAU-specific with general pain; and (c) TUD-specific with musculoskeletal pain. A significant enrichment indicates that the parameter shows greater expression of an annotation than is proportionally expected. The red asterisks indicate significance at a false discovery rate (FDR)-corrected P < 0.05. The error bars are the 95% CIs. Details on all enrichments can be found in Appendix S2. PAU = problematic alcohol use; PI genes = protein-truncating invariant genes; TUD, tobacco use disorder.

liability. Smoking may contribute to pain chronification by strengthening reward circuits [21], though other evidence suggests its relationship with chronic pain operates entirely through genetic pathways and potential gene–environment correlations, rather through a direct causal path [56]. Taken together, the associations we report may reflect a heritable predisposition through which chronic pain and prolonged substance use influence each other, partially through a predisposition to maladaptive neurobiological plasticity.

Substance-specific associations

General chronic pain shared a positive genetic correlation with PAU, consistent with past literature [57]. However, upon controlling for addiction variance, there remained a significant negative association between general chronic pain and PAU. This pattern likely reflects a suppression effect, where the addiction factor captures the broad, shared liability that underlies both traits. The remaining variance that inversely relates to pain liability may potentially reflect protective behavioral mechanisms, such as the avoidance of alcohol owing to pain medication interactions.

Furthermore, controlling for both addiction and PAU-specific variance, general chronic pain shared an additional negative association with alcohol use. Alcohol consumption may exacerbate some pain conditions, such as migraine [58]. This exacerbation may cause a decrease in alcohol consumption at the population level, and show up as a negative genetic association. A Mendelian randomization study found evidence of reverse causality of this type, that is, genes associated with migraine cause reduced alcohol consumption [59], whereas alcohol use disorder shows bidirectional positive associations with multi-site chronic pain [14]. Thus, our finding of a negative correlation between chronic pain and alcohol consumption may be indicative of a causal relationship between them. Taken together, these findings emphasize the importance of partitioning substance-use variance when examining genetic associations with pain, as shared and substance-specific liabilities may act in opposing directions.

Even though cigarette smoking frequency is considered a good genetic proxy of nicotine dependence [60], we still found unique significant genetic variance between general pain and tobacco use when controlling for addiction and TUD. Beyond a shared risk for maladaptive plasticity in reward processing, smoking frequency may affect pain by degrading end-organ integrity [11]. Synthesizing our findings, there are unique pathways between different smoking phenotypes and pain conditions, and questions of causality versus shared pleiotropy may not be generalizable to the umbrella phenotypes of chronic pain and tobacco use.

Musculoskeletal pain showed a distinct pattern of relationships with SUDs, compared with general chronic pain. Musculoskeletal pain was not genetically related to common addiction but had a substance-specific association with TUD. Tobacco smoking may have a causal effect on musculoskeletal systems specifically. Notably, the general pain factor partitioned out all central nervous system pathways, and the musculoskeletal pain factor captures pathways specific to the musculoskeletal system. There is robust evidence of smoking having adverse effects on the integrity of muscles, tendons, cartilage and ligaments, which may be why we find a TUD association but not an addiction association [61]. As previously mentioned, there is conflicting evidence on how tobacco and pain are related [14, 55, 56]. Multiple candidate gene-interaction studies, the findings of which should be interpreted cautiously [62], have found that certain genotypes have exacerbated biomarkers of musculoskeletal disease in those who smoke [61]. Hence, smoking may have a direct impact on musculoskeletal disease, while also creating magnified effects in those with a genetic predisposition.

When controlling for addiction and PAU, musculoskeletal pain shared a positive genetic correlation with alcohol use-specific variance, potentially owing to vertical pleiotropy driven by self-medication for musculoskeletal-specific pain. Unlike migraines, where alcohol use can trigger an episode and discourage drinking [58], alcohol may provide temporary pain relief to those with musculoskeletal pain conditions, leading to negative reinforcement [19]. In an animal model, alcohol consumption deteriorated multiple musculoskeletal systems [63]. A systematic review has shown robust evidence of alcohol consumption degrading skeletal systems in humans [64], which may be why we find a positive genetic correlation between alcohol use and musculoskeletal pain. Alternatively, alcohol use, compared with problematic alcohol use, has gene sets specific to the cellular responses to drinking; musculoskeletal pain and alcohol use may both implicate a heritable susceptibility to chronic inflammation [23].

Neither pain factor had specific genetic correlations with CanUD or OUD after controlling for associations with the addiction factor. Prior to controlling for addiction, the genetic correlation between general pain and OUD was the highest of all the SUDs, which is consistent with past literature [57]; however, this association was fully captured by the association with the addiction factor. The OUD–pain co-occurrence may function entirely through shared, central mechanisms common to addiction, such as a predisposition to aberrant reward processing. The lack of significant association with CanUD controlling for addiction risk suggests that shared genetic risk between general pain and CanUD also acts through common addiction risk.

Future directions

While this study distinguishes between common addiction liability and substance-specific effects on pain, the causal direction between chronic pain and substance use remains speculative. Future research can apply Mendelian randomization on these partitioned effects to probe causal direction. In addition, identifying neurobiological mediators of this shared genetic liability represents a critical next step. Large-scale imaging genetics resources (e.g. UK Biobank) now provide extensive structural and functional magnetic resonance imaging (MRI) phenotypes [65] that could be used to clarify whether alterations in reward- and affect-related circuits mediate genetic overlap between these conditions. Integrating these data through GenomicSEM mediation could distinguish whether shared liability reflects common genetically driven neural architecture or downstream effects of pain and substance use. Multi-modal imaging measures (e.g. cortical thickness, white matter integrity, resting-state connectivity) may also help link the cell type-specific enrichments identified here to systems-level pathways. Such work could yield mechanistic insights into how genetic predispositions manifest in brain circuitry and inform targeted interventions for co-occurring pain and addiction.

Limitations

The GWASs require large sample sizes, and the traits are often meta-analyzed across multiple cohorts. As a result, there was some incongruency with controls, with some controls being unscreened for SUDs and/or unexposed to the substances. These design decisions in the original GWASs likely added noise to our variables. Latent variable modeling extracts common variance and leaves residual variance of each indicator and its noise. Hence, the factor-level correlations may not be as impacted by this issue. However, when allowing the general pain factor to associate with CanUD, we may not have identified genetic associations with the significant CanUD residual, owing to it being a noisier trait rather than there being a true lack of association. Consortiums and biobanks are collecting and releasing updated data with higher powered, less noisy traits. Furthermore, the partitioning of problematic use/consumption for cannabis and opioids was not done, as there are no publicly available, well-powered GWASs on cannabis and opioid frequency yet published.

These analyses were conducted using GWASs of European-like ancestry and may not be generalizable to non-European-like ancestries (e.g. African-like or East Asian-like ancestries). GWASs stratify by ancestry to prevent false positives for trait associations that also correlate with ancestry [66]. There is a pervasive lack of data in diverse ancestries, which currently prevents us from replicating our questions of interest in other ancestries [67].

CONCLUSION

Genetic risk common to 24 chronic pain conditions shares substantial overlap with addiction risk common to four substances. Chronic pain liability shares a negative, substance-specific association with problematic alcohol use, and differential relationships with alcohol and tobacco use. Musculoskeletal pain, common to 11 musculoskeletal conditions, does not relate to addiction risk, but instead has its own risk profile with TUD. There are both common and unique biological pathways that underlie pain and substance use co-occurrence. Functional enrichment of multiple neurological subtypes suggests potential biomarkers, shared mechanisms and clinical targets.

AUTHOR CONTRIBUTIONS

Lydia Rader: Conceptualization (lead); data curation (lead); formal analysis (lead); methodology (lead); visualization (lead); writing—original draft (lead); writing—review and editing (lead). **Katerina Zorina-Lichtenwalter:** Data curation (supporting); methodology (supporting); writing—review and editing (equal). **Alexander S. Hatoum:** Data curation (supporting); methodology (supporting); writing—review and editing (equal). **Michael C. Stallings:** Conceptualization (supporting); methodology (supporting); writing—review and editing (equal). **Tor D. Wager:** Funding acquisition (equal); writing—review and editing (equal). **Naomi P. Friedman:** Conceptualization (equal); funding acquisition (lead); methodology (equal); supervision (lead); writing—review and editing (equal).

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DECLARATION OF INTERESTS

The authors do not have any conflicts of interest to disclose. An earlier version of this study was completed as partial requirement for a doctoral dissertation.

DATA AVAILABILITY STATEMENT

Data pre-processing and analysis code are publicly available on GitHub (https://github.com/lydia-rader/sud_pain_gsem_2025).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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